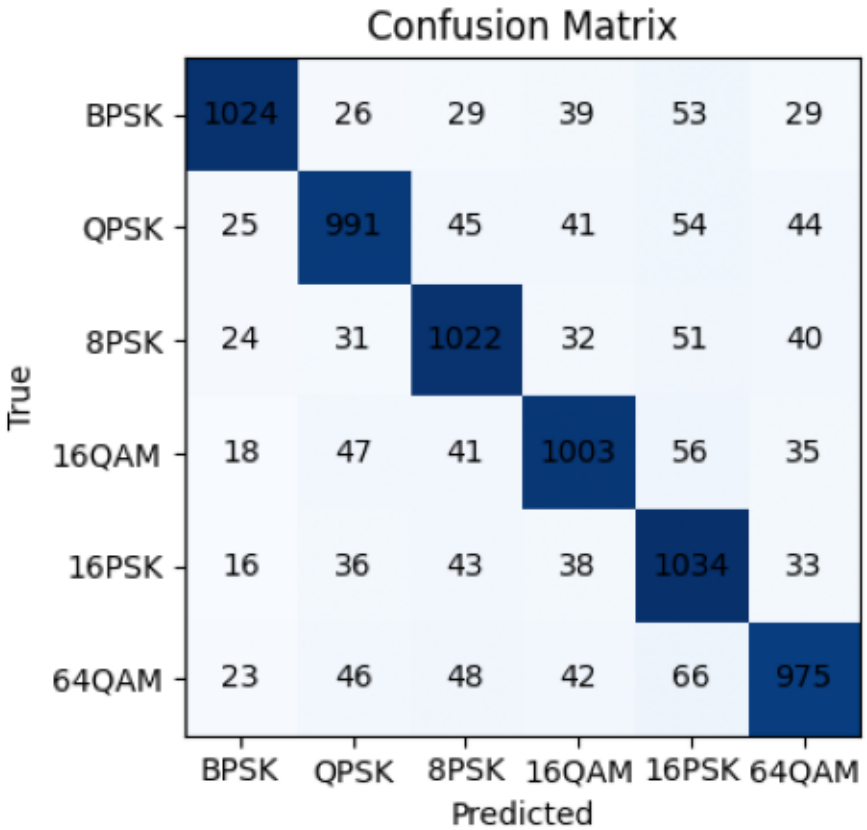
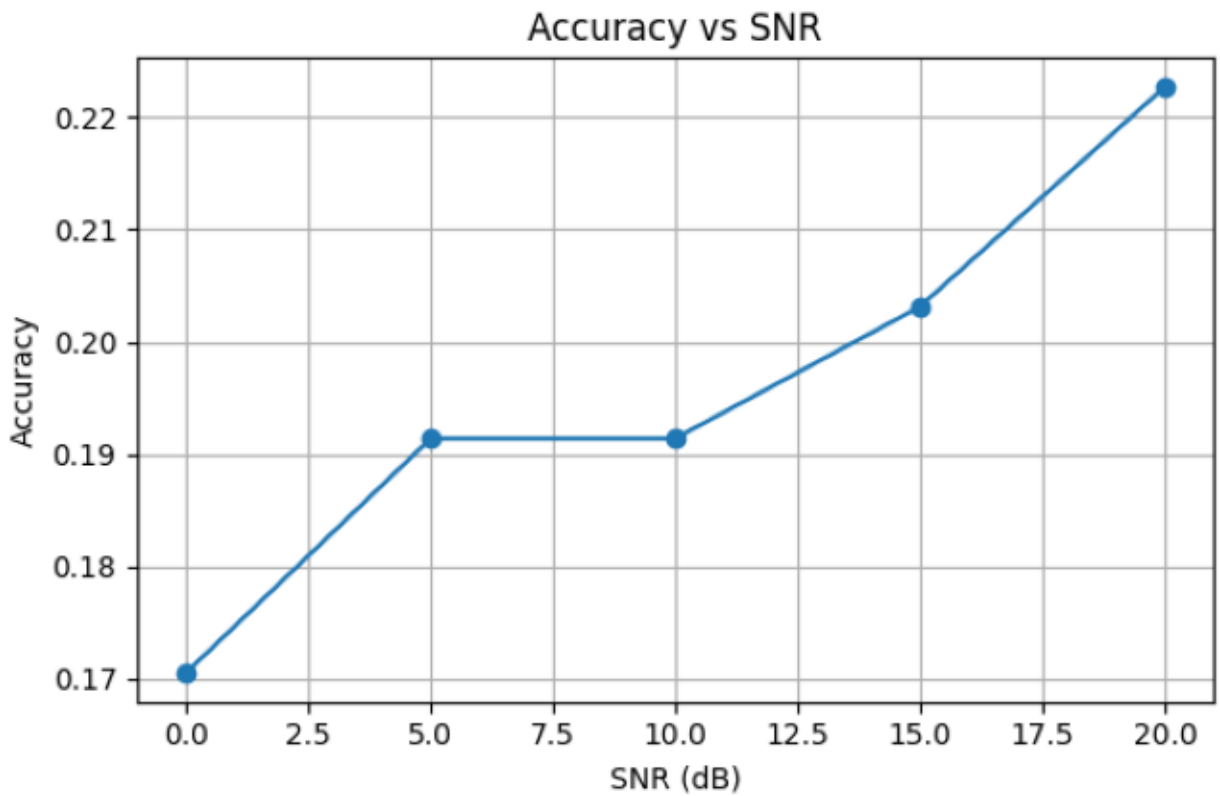
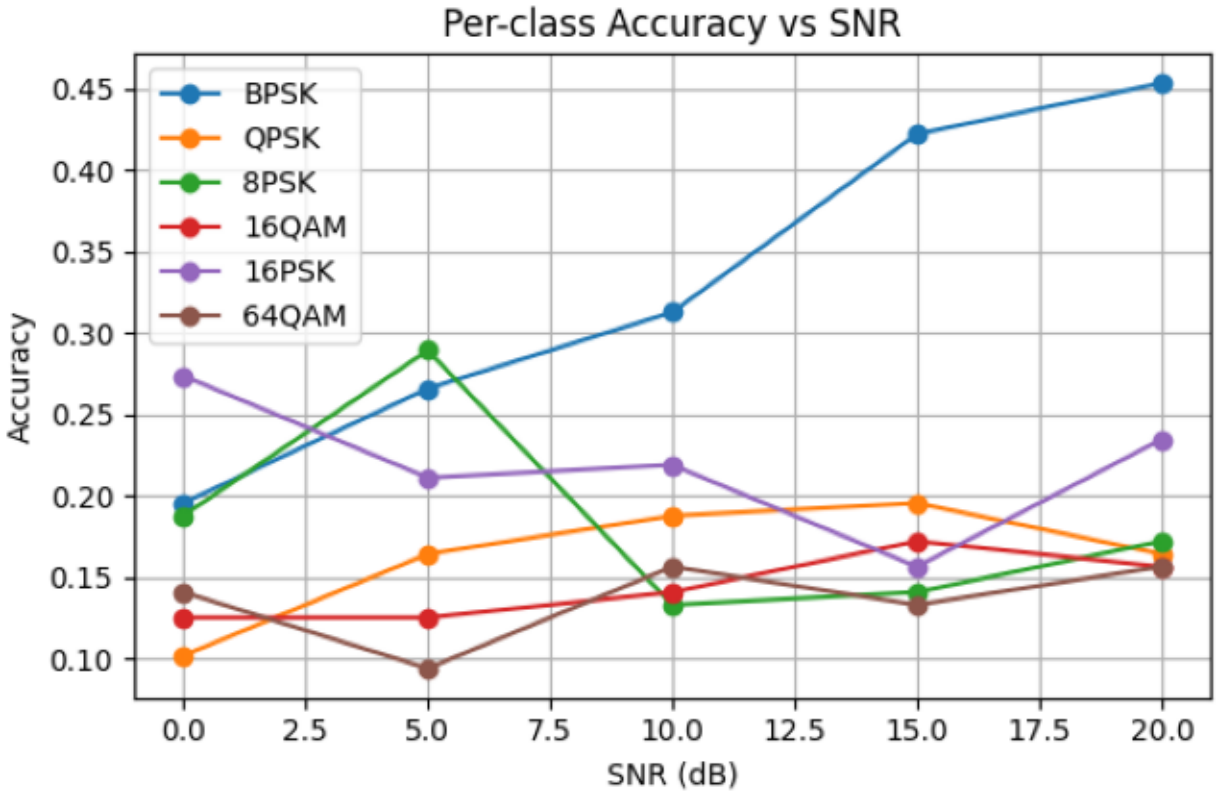
Neuromorphic Cognitive Radio

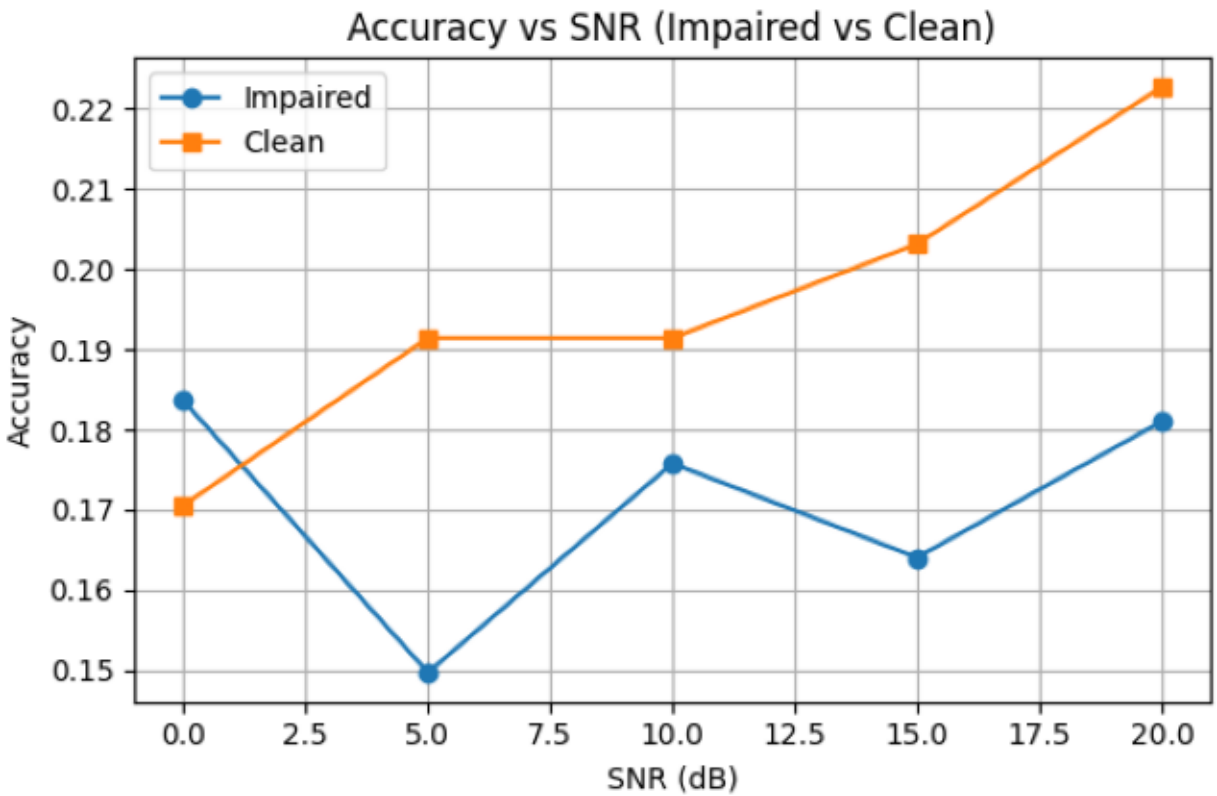
Summary

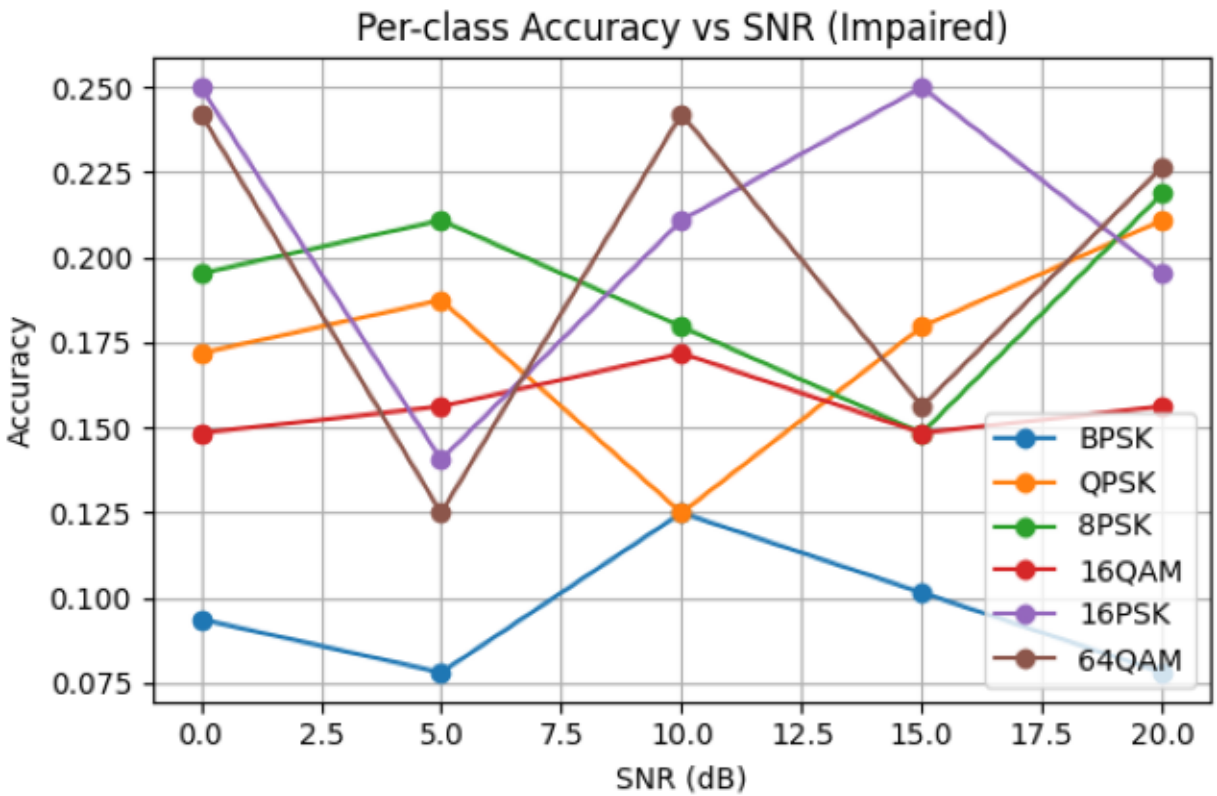
Accuracy: 0.8401
Total Spikes (hidden): 714076
Estimated Energy per inference (nJ): 2856.30
Intel Loihi: bits=9, cores=128, neurons/core=1024, est_energy_nJ=3213.34
BrainChip Akida: bits=4, cores=384, neurons/core=4096, est_energy_nJ=1428.15
SpiNNaker: bits=16, cores=864, neurons/core=1024, est_energy_nJ=5712.61
Per-class metrics (precision, recall, F1):
BPSK: P=0.906, R=0.853, F1=0.879
QPSK: P=0.842, R=0.826, F1=0.834
8PSK: P=0.832, R=0.852, F1=0.842
16QAM: P=0.839, R=0.836, F1=0.838
16PSK: P=0.787, R=0.862, F1=0.823
64QAM: P=0.843, R=0.812, F1=0.828
Macro-F1: 0.840

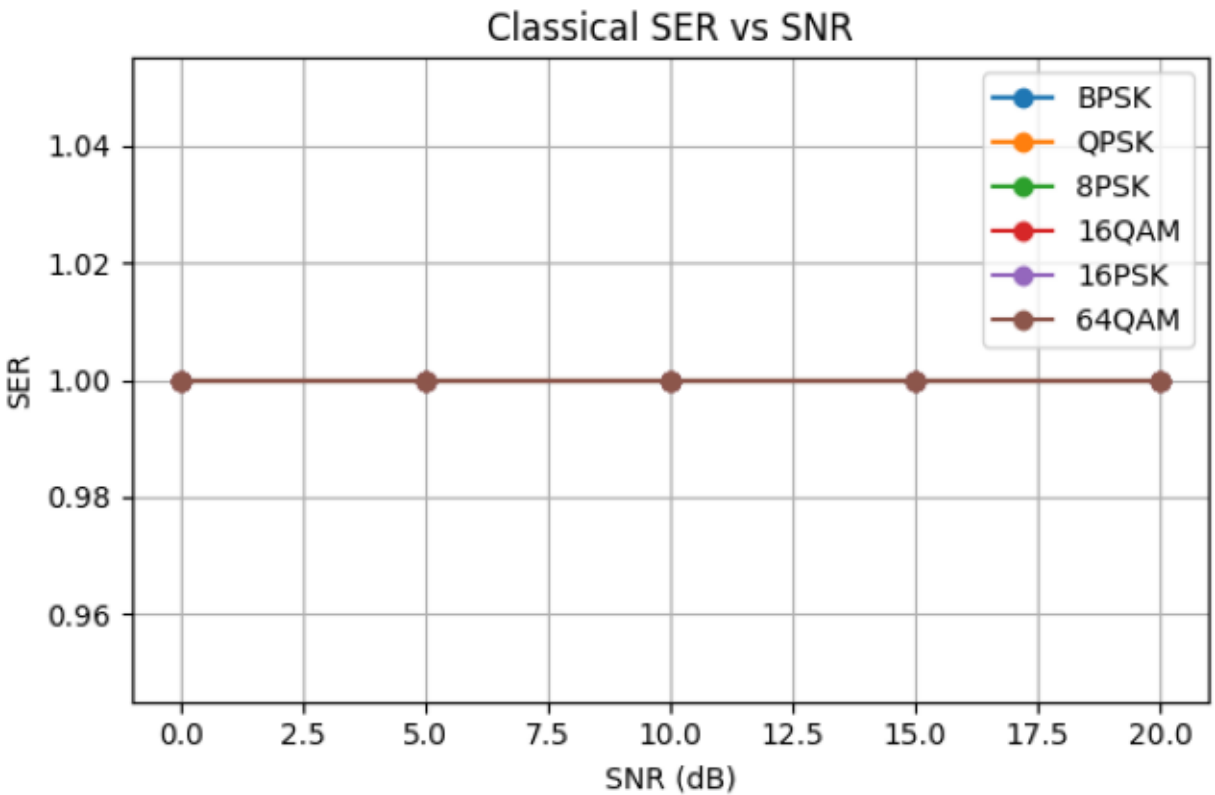

Confusion Matrix
True
Predicted
BPSK
QPSK
8PSK
16QAM
16PSK
64QAM
1024 26 29 39 53 29
25 991 45 41 54 44
24 31 1022 32 51 40
18 47 41 1003 56 35
16 36 43 38 1034 33
23 46 48 42 66 975

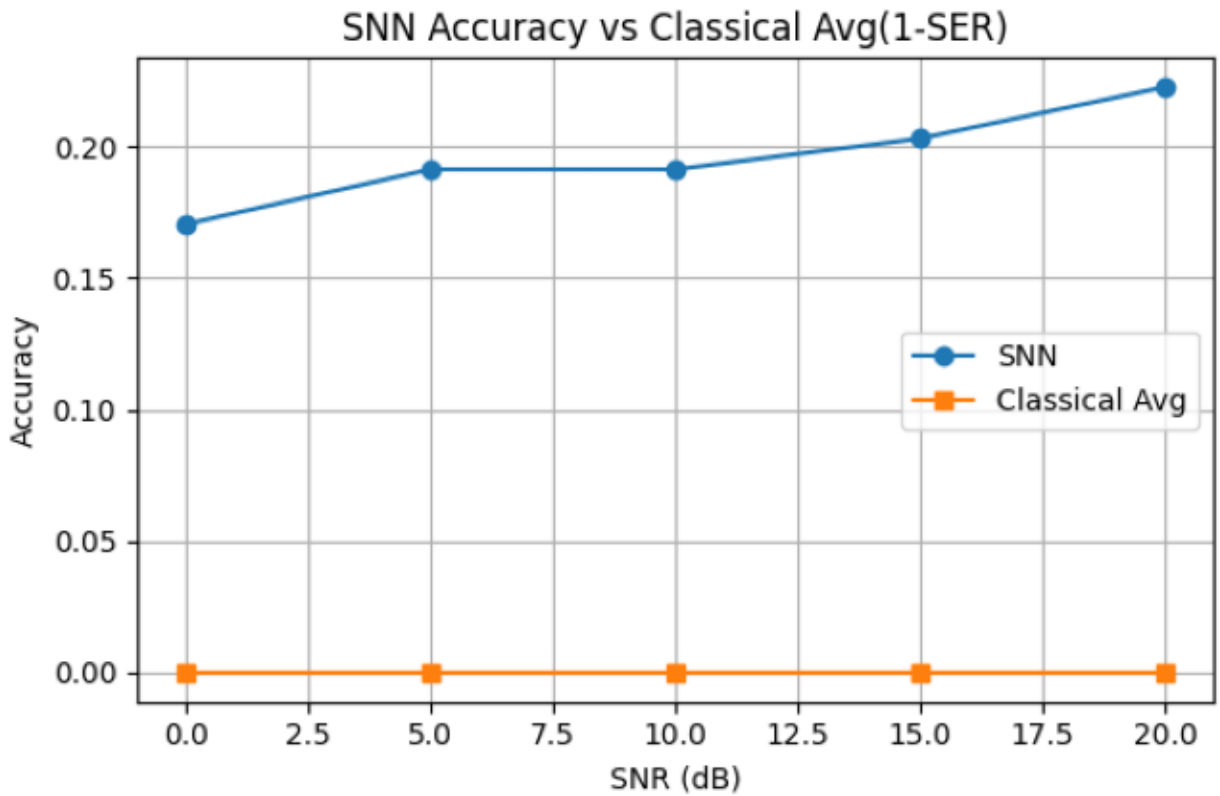

Accuracy vs SNR
Accuracy
0.22
0.21
0.20
0.19
0.18
0.17
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
SNR (dB)


Per-class Accuracy vs SNR
BPSK
QPSK
8PSK
16QAM
16PSK
64QAM
Accuracy
0.45
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
SNR (dB)


Accuracy vs SNR (Impaired vs Clean)
Impaired
Clean
Accuracy
0.22
0.21
0.20
0.19
0.18
0.17
0.16
0.15
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
SNR (dB)


Per-class Accuracy vs SNR (Impaired)
Accuracy
SNR (dB)
0.250
0.225
0.200
0.175
0.150
0.125
0.100
0.075
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
BPSK
QPSK
8PSK
16QAM
16PSK
64QAM


Classical SER vs SNR
BPSK
QPSK
8PSK
16QAM
16PSK
64QAM
1.04
1.02
1.00
0.98
0.96
SER
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
SNR (dB)


SNN Accuracy vs Classical Avg(1-SER)
Accuracy
0.20
0.15
0.10
0.05
0.00
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
SNR (dB)
SNN
Classical Avg